An International Comparison of Equity in Education Systems

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An international comparison of equity in education systems

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This paper uses pupil responses to the PISA study in 2000 for all EU countries. Using indicators of the pupil intakes to schools and their outcomes it computes segregation indices for 15 countries, and then tries to explain the resulting patterns in terms of the characteristics of national school systems. Segregation by sex in each country is explicable by its provision of single-sex schools, religious schools, and the use of academic selection in allocating school places. Segregation by outcome is largely explicable by the use of academic (and other forms of) selection. Segregation by parental occupation or country of birth is lower in countries allocating places at school through elements of choice or with relatively little governmental control of schools rather than use of rigid catchment areas or selection. In all countries there are small gaps between the performance of boys and girls in reading, in favour of girls. This gap is generally smaller in countries with the highest overall scores. Overall, the Scandinavian countries of Sweden, Finland and Denmark show less segregation on all indicators, while Germany, Greece and Belgium show the most. The UK has below average segregation in terms of all indicators except sex, despite a commonly held but unfounded view that segregation in the UK is among the worst in the world.

Introduction

This paper is based on a secondary analysis of the Programme for International Student Assessment (PISA) data from 2000, as part of our Socrates-funded study of equity in education systems across the EU. From these results, and those of our own large-scale survey of schools and teachers in five EU countries, we are constructing EU-wide indicators of equity and inequality for annual publication.

Comparison between different national education systems is difficult for many reasons, but largely because school systems differ in so many respects that it is difficult to mount a convincing argument that any one difference is related to another (Gorard, 2001). Countries such as Germany and Austria offer a choice—frequently linked to ability—between vocational, technical and general education at the age of 12 (in the case of Germany) and 14 (in the case of Austria). Other countries, such as the UK, Denmark, Finland and Portugal, offer a general education up to the end of compulsory schooling (usually 16). The UK is unique among the European Union countries in employing a rigorous regime of prescriptive curriculum and centralized control of national testing. No other country offers

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national tests at each phase of compulsory education. Indeed, countries such as Austria, the French and Flemish communities of Belgium, Spain, Italy, Germany and Luxembourg offer no formal external testing at all during compulsory education (Eurydice, 2002). In these countries assessment is frequently formative and teacher-led. Other countries such as Denmark, Greece, Portugal, Sweden, Ireland and the Netherlands offer examinations at the end of compulsory education, usually leading to a school-leaving certificate.

Despite their limitations, such comparisons can be very useful. They allow researchers and policy-makers in one country to put the position of their own country in perspective. For example, according to a leader column in the UK *Times Educational Supplement* in June 2002,

> As every international comparison has shown, English schools are more socially differentiated than *any* others in Europe. Some hardly warrant the description 'comprehensive' at all, thanks to the parental choice policies pursued by successive governments. They may be even more socially stratified than the old grammar and secondary moderns they replaced. (p. 20)

Claims such as these are quite common, and contribute to what has become a 'crisis account' of the state of the UK education system and its schools. However, these claims are not based on the kind of evidence that the PISA study can provide.

International comparisons also add another dimension to the testing of explanatory causal models. Many developed countries are concerned to understand the impact of choice policies on patterns of attainment and social segregation. Looking at patterns of change in one country over time before and after the introduction of choice policies gives one view of this impact (Gorard *et al.*, 2001a). Looking at differences in segregation between countries, some of which have introduced choice policies and some which have not, gives another 'triangulating' view. This is especially important for the UK at present since the distinctions between choice, diversity and independence from government control have become confused by policy-makers and some analysts. All of these issues, and that of underachieving boys, are addressed by the evidence presented in this paper.

Education is increasingly seen as something the State owes to its citizens, and it is the State's responsibility to ensure it provides an equitable education for all. As a consequence, ensuring a fair and equitable education system has political as well as social implications. Schools have a responsibility to society for producing at least minimally qualified individuals, but there are diverse ways in which inequalities can manifest themselves within schools, a particular example being the experiences of the poorest pupils. For example, some pupils achieve better results than others, attend more 'effective' schools or have longer school careers (Meuret, 2001). Most research has shown that the poorest pupils are the least successful in school, in terms of examination performance (Heath, 1989). In the UK, as the academic achievement of all groups of pupils has improved, certain levels of qualification, for example, the General Certificate of Secondary Education (GCSE) Grade G have become almost universal. The near saturation at this relatively low level of certification has arisen in part because pupils who have come predominantly from the working class have caught up with their higher socio-economic status peers in this regard (Raftery & Hout, 1993). But it is also as a consequence of the proportionate reduction in the size of the working class itself. One consequence of this has been that the qualifications required by the labour market, and part of the 'key' to transition between the social classes, have moved up towards A level. This shift of focus from secondary- to tertiary-level education also has implications for students from the poorest homes if for economic reasons they might be less able to remain in education beyond the age of 16.

It is clear that a fair and equitable education system is important for many reasons—social, political and economic, but how does this relate to the desire to raise academic standards? Growth in standards (like economic growth) may be related to growth in inequality (at least temporarily). According to some researchers, rather than widening their access, schools have become more stratified in terms of attendance and achievement between different groups (for example, according to gender, ethnicity and social group) and hence have become less equitable (Gewirtz *et al.*, 1995). Despite the lack of solid evidence for these claims, resultant dissatisfaction with the education system has contributed to the 'crisis account' of the state of British schools. Some of the evidence for this apparent 'underachievement' of pupils in British schools has come from the results of international standardized tests. These have been reported as providing evidence of a 'two-track' education and training system in the UK, where the top 10% of the school population perform relatively well, but where the less able are represented by a long tail of underperformance (Reynolds & Farrell, 1996; DfES, 2001). Comparisons such as these have led to calls for policy borrowing from more academically successful nations and also for an increase in initiatives to raise academic standards.

The recent education White Paper in England, *Schools—achieving success*, called for increased autonomy and diversity in secondary schools so that the nation can build a world-class education system and 'transform the knowledge and skills of its population' (DfES, 2001, p. 7). Among the proposed changes are more difficult National Curriculum tests for 11-year-olds (Henry, 2002) and an increase in the number of faith and specialist schools—schools that according to government research have a 'proven success in raising academic standards' (DfES, 2001, p. 41). According to Morris (DfES, 2002, p. 6), 'the model of comprehensive schooling that grew up in the 1960s and 1970s is simply inadequate for today's needs ... the keys are diversity not uniformity'. The then Minister therefore proposed to convert the comprehensive system into a tiered ladder of diverse schools, including 33 new academies, 300 advanced schools and 2000 specialist schools in England. These are in addition to an expansion of faith-based schools in England and Welsh-medium schools in Wales, and the continued existence of fee-paying and foundation schools. What can the PISA study tell us about the likely outcome of such policies?

Methods

The results presented here are taken from the PISA study, which was conducted in 2000 (OECD, 2000). The study covered 265,000 15-year-old students from 32 countries, but this paper focuses on the 74,356 pupils in 2647 schools in the 15 countries of the EU. The survey items included tests in literacy, maths and science, and pupil and school questionnaires on aspects of student motivation, use of ICT, school organization and so on. Complete coverage of performance in the tests is given for literacy only, and there are plans to extend the coverage for maths and science in subsequent surveys. This paper reports the results of the analyses for all 15 EU countries. Results for the Flemish- and French-speaking communities of Belgium, and England, Scotland and Northern Ireland are presented separately in Smith and Gorard (2002).

Indicators of relative disadvantage

The primary aim of this study is to consider the distribution of groups of pupils in schools. In particular we are concerned with the distribution of those who fall into the most disadvantaged categories in terms of poverty and other factors associated with educational achievement. Where appropriate, the analysis is conducted for pupils who fall into the lowest 10% for a particular indicator (such as family income), which means that they are considered to be disadvantaged relative to their peers. Otherwise the pupils appear in one of two categories (such as being born in the test country or not). The relative segregation of this group in schools in different EU countries is then estimated using a variety of segregation indices, described below.

Indicators for which segregation values have been calculated

Parental occupation. Students were required to report their mothers' and fathers' occupations, which were classified according to International Standard and Classification of Occupations (ISCO) criteria and transferred to a scale, as described by Ganzeboom and Treiman (1996). The ensuing variable is based on either the father's or mother's occupation, whichever is higher.

Family wealth. The PISA index of family wealth is based on the students' responses to questionnaire items on: (i) the availability in their own home of a dishwasher, a room of their own, educational software and a link to the Internet; and (ii) the number of mobile phones, television sets, computers, cars and bathrooms they have at home.

Pupil's country of origin. The student questionnaire asked students to report in what country they and their parents were born. Answers indicated either that the individual was born in the country in which the test was being taken, or another country. Those born in another country are the minority group.

Performance in reading examination. PISA gives two kinds of estimate for performance in reading, maths and science—a weighted likelihood estimate and a set of 'plausible' values. The plausible values are not test scores; instead they are designed to give good estimates of parameters of student populations, rather than estimates of individual student proficiency. As a result, the plausible values are better suited for describing the performance of a population. In this exploratory analysis, one set of plausible values is examined.

Sex. The student questionnaire also asked for their sex. It makes no difference which category is used as the 'minority' group, and here we used female.

Determining the unit of analysis

For continuous measures the more disadvantaged groups were estimated by taking the lowest 10% of the population. However, there are different methods of determining these groups of students. For example, the lowest 10% of a particular indicator can be ascertained in one of two ways—first by estimating the lowest 10% over the entire EU population (type E) or alternatively by the lowest 10% over each country separately (type C). Unsurprisingly, both methods yield somewhat different figures.

Within the UK, 1087 pupils have parents whose occupation places them in the lowest 10% of the population. However, if the lowest 10% group is defined over the entire EU, 399 of these pupils are no longer included in the analysis, presumably because wealth in England is higher than the EU average. This in turn affects calculation of the segregation values. For some other countries the reverse happens, and pupils who would otherwise be excluded in a Europe-wide analysis (Type E), are included in the disadvantaged groups by country. One example is Portugal where there are 294 additional pupils in the disadvantaged group (lowest 10%) if the group is estimated over the entire EU sample. The results for each indicator taken over both the EU and country have been calculated, but the most useful unit of analysis for national comparisons is usually taken to be the country level (Type C). Type E comparisons would be more appropriate for considering changes in EU-wide segregation over time. The full set of results for both types, and all segregation indices appear in Smith and Gorard (2002).

Segregation indices used in the analysis

The standard method of analysing the distribution of a range of individual characteristics used here is the segregation index (S). This index is defined as the 'proportion of disadvantaged students who would have to exchange schools within the area of the analysis, for there to be an even distribution of this group among the population'. S has been shown to give similar results to other indices of the same type (Gorard & Taylor, 2002). For comparative purposes, the figures have also been calculated for two other widely used measures of segregation: the Dissimilarity index (D) and the Gini Coefficient (G).[1] In addition, we have prepared Lorenz curves for each indicator but these curves intersect to such an extent that they are practically useless as discriminators.

Table 1. Standard table of segregation

	Number of pupils in minority group	Number of pupils in majority group	Total
$School_1$	A_1	B_1	C_1
$School_2$	A_2	B_2	C_2
.....	.....	.....	....
$School_n$	A_n	B_n	C_n
Total	X	Y	Z

In this study, individuals were designated as belonging to either the minority group (for example those in the lowest ability 10% or those born outside the test country) or the majority group (for example, the wealthiest 90%). The situation is summarized in Table 1.

The segregation index

$$S = 0.5 \times \sum \left| \left(\frac{A_i}{X}\right) - \left(\frac{C_i}{Z}\right) \right|$$

Here A_i and C_i represent the number of pupils in the minority group (for example, the lowest 10% or those born outside test country) and total number of pupils in each school respectively. X and Z are the total number of pupils in the minority group and the total number of all pupils sampled in each country.

There is a great deal of discussion in the academic literature as to which segregation index is the most effective (see Massey & Denton, 1988). In this study segregation values were calculated for each of the three main indices, and the relationship between each pair of the indices was considered in an attempt to find the most appropriate index. The correlation between each pair was very high (from 0.95 to 1.00 Pearson's R). Using country of birth as the indicator and country as the level of analysis, for example, the correlation between S and D was 1.00 and between S and G was 0.98. Thus, in many respects it may not matter which index is used. However, S has been shown to be the only one which is entirely unaffected by differences in the absolute values of any indicator between countries. Further, in factor analyses, S has the highest loading onto the underlying variable that all indices seek to describe. Finally, S is the arithmetic equivalent, but using categorized variables, to the mean deviation, meaning that S is easier to interpret than others. For these reasons S is used throughout as the standard comparator. It is important to note that despite the large sample in the study as a whole, when we consider a rare characteristic, such as those born outside the test country, then the focus is on a relatively small number of cases. Small differences either within or between indices are therefore not treated as substantive in the analysis described here.

Table 2. Segregation by all indicators, type C

Country	Parental occupation	Family wealth	Reading score	Country of origin	Sex
All EU	33	28	49	48	15
Austria	36 (+.04)	24 (−.08)	62 (+.12)	49 (+.01)	28 (+.30)
Belgium	36 (+.04)	26 (−.04)	66 (+.15)	45 (−.03)	22 (+.19)
Denmark	33	28	39 (−.11)	42 (−.07)	10 (−.20)
Finland	36 (+.04)	21 (−.14)	27 (−.29)	55 (+.07)	7 (−.36)
France	31 (−.03)	31 (+.05)	56 (+.07)	47 (−.01)	12 (−.11)
Germany	36 (+.04)	33 (+.08)	61 (+.11)	41 (−.08)	11 (−.15)
Greece	43 (+.13)	26 (−.04)	58 (+.08)	48	13 (−.07)
Ireland	29 (−.06)	30 (+.03)	39 (−.11)	45 (−.03)	30 (+.33)
Italy	30 (−.05)	27 (−.02)	58 (+.08)	55 (+.07)	23 (+.21)
Luxembourg	24 (−.16)	23 (−.10)	41 (−.09)	24 (−.33)	12 (−.11)
Netherlands	30 (−.05)	23 (−.10)	66 (+.15)	41 (−.08)	10 (−.20)
Portugal	40 (+.10)	36 (+.13)	48 (−.01)	35 (−.16)	8 (−.30)
Spain	32 (−.02)	28	40 (−.10)	57 (+.09)	10 (−.20)
Sweden	27 (−.10)	29 (+.02)	29 (−.26)	40 (−.07)	9 (−.25)
UK	31 (−.03)	26 (−.04)	43 (−.07)	46 (−.02)	16 (+.03)

Results

Table 2 shows the segregation values for all countries on all indicators, followed, in brackets, by the proportionate deviation between each value and the value for the EU as a whole. Table 3 summarizes the characteristics of each national school system. The first column shows the percentage of pupils in academically selective schools, the second shows the percentage of pupils in single-sex schools, the third shows the percentage of pupils in religious-basis schools, and the fourth shows the percentage of pupils in schools fully controlled by the government. The fifth gives an indication of the overriding system of school allocation (tiered selective schools, strict catchment areas, some elements of choice, and national choice systems). The results are discussed first country by country, before a summary is attempted.

Austria has a generally segregated school system on all indicators except family wealth, and is highly segregated by sex. It is notable that segregation by family wealth does not correlate highly with segregation by parental occupation across the EU. Since family wealth is a totally artificial composite variable, more significance is attached here to occupational figures. Austria has 10% of the pupils in the survey in single-sex schools, but this figure alone cannot explain the very high level of segregation by sex. The UK, for example, has more pupils in single-sex schools but less segregation by sex. Overall, sex segregation in Austria is most easily explained by the high level of academic selection. Since girls generally do better than boys in examination performance in all developed countries, academic selection can also lead to segregation by sex even where schools are not single-sex in nature (but not always; see comments on Germany). Even though there are elements of choice, these will be inhibited by the high level of selection.

Belgium is only slightly less segregated than Austria, perhaps because it uses religious selection (33%) as well as academic selection (25%). In fact the total proportion of pupils in selective schools in the two countries is remarkably similar. Therefore, although most schools are not government-controlled and there is free choice, the differentiation of schools by religion is linked to considerable segregation by outcomes and sex. Denmark has around EU-average segregation (or below) on all indicators. The fact that there are no single-sex schools leads to low segregation by sex. The fact that there is little selection of any kind means that segregation by outcomes is low despite a widespread use of catchment areas.

Finland has almost no selection, and free choice of government-controlled schools. This leads to a system with little segregation, except by country of origin (and this anomaly is worthy of further investigation), and very little segregation in terms of examination outcomes. France, like Denmark, has around EU-average segregation. It has a reasonable proportion of academically selective schools leading to above-average segregation by outcomes, but not by sex or religion. Germany has a tiered system with academic selection, and a consequently high level of segregation by outcome (as well as by family background). Here, the lack of single-sex schools is related to little segregation by sex.

Greece has a basic catchment area system leading to segregation by occupation and outcome. The difference to Denmark is striking and may be explained by a lower population density and a different housing mix. In other words, a catchment area system is more divisive when there is further to travel or where local housing tends to be of one kind and price range. Ireland has a high level of segregation by sex stemming from a high proportion of pupils in single-sex schools (44%), and in religious schools (31%). Ireland has relatively low segregation on other indicators, and a system of free choice among schools largely free of government control.

Italy has a catchment area system of allocation to government-controlled schools, and above-average segregation by outcomes. It also has a very high level of segregation by sex given the proportion of pupils in designated single-sex schools. Luxembourg has low segregation on all indicators. This is somewhat surprising, given the 41% of pupils in academically selective schools.

The Netherlands has a very selective (79%) and tiered system leading to highly segregated outcomes. However, most schools are not government-controlled and in other respects the system is equitable. Portugal has a catchment system and government-controlled schools, and high levels of segregation by family background. Spain also has a catchment system but less government control and considerably lower of levels of segregation by family than Portugal. Sweden has no selection, elements of choice, and very low levels of segregation by intake and outcome. The UK has a national system of choice, but significant elements of selection, leading to around EU-average levels of segregation (or just below).

It is difficult to summarize these findings across 15 countries accurately, but we can make a few simplifying assumptions. First: parental wealth is not a useful indicator, because it does not correlate highly with any other (even parental occupation). Second: segregation in schools by sex is largely a result of single-sex provision ($r = 0.69$), sometimes coupled with academic and religious selection.

Table 3. Summary characteristics of national school systems

Country	Academic selection%	Single-sex%	Religious selection%	Govnt Control%	Choice policy
Austria	52	10	8	90	Some
Belgium	25	6	33	27	Free choice
Denmark	4		16	76	Catchment
Finland	1		4	96	Free choice
France	26	1	9	77	Catchment
Germany	31	3	6	96	Tiered
Greece	5	4	2	97	Catchment
Ireland	8	44	31	39	Free choice
Italy		6		94	Catchment
Luxembourg	41	14	19	86	Some
Netherlands	79	7	17	26	Tiered
Portugal	5	1	7	93	Catchment
Spain	1	2	13	63	Catchment
Sweden	1		2	96	Some
UK	19	16	17	95	Free choice

Third: segregation by parental occupation and by country of origin do *not* correlate consistently or highly with any other measure. They are, presumably, primarily measures of how mixed the intakes to schools are as a result of admission policies other than selection, and will also be a function of population density and mobility and the nature of housing.

Segregation by outcome is strongly related to the proportion of academic selection in each country ($r = 0.61$). Therefore, it would be safe to conclude that selection, and not social segregation as such, is responsible for the largest gaps in attainment. In a national comprehensive school system there will be geographically and economically driven social segregation, but this is very different from a system of overt selection with tiers of schools specially designed for different ability ranges. Religious-based schools do not have a strong relationship with segregation by outcome, but are clearly linked to both single-sex provision (0.61) and lack of government control (-0.79). All other things being equal, systems allowing parents a choice of schools (that is, without tiering or selection) have lower intake segregation and, especially, outcome segregation.

Table 4 presents the results for reading performance according to the pupils' score on the PISA indicator of wealth. The pattern is the same when parental occupation is considered—pupils who fall into the lowest 10% of either category perform less well on the reading tests. A similar pattern emerges for pupils' country of origin; pupils who were born outside the test country were more likely to achieve lower scores on the reading test. In general, countries with the lowest gap in reading performance between richest and poorest are also those that have relatively high scores, even for the poorest 10%. Finland, Ireland and the Netherlands have very high scores for both groups, while France, Germany and Luxembourg, with heavily selective systems, have both very low scores for the poorest 10% and only average

Table 4. Mean reading score according to PISA indicator of family wealth

Country	Poorest 10%	Richest 90%	Difference
Austria	477	502	2.6
Belgium	489	519	3.0
Denmark	479	502	2.3
Finland	540	550	0.9
France	465	509	4.5
Germany	454	504	5.2
Greece	456	475	2.0
Ireland	512	530	1.7
Italy	472	492	2.1
Luxembourg	385	452	8.0
Netherlands	541	543	0.2
Portugal	422	483	6.7
Spain	469	499	3.1
Sweden	495	519	2.4
UK	502	529	2.6

scores for the richest 90%. The UK has the fourth highest score for the poorest 10% and the third highest score for the richest 90%.

Pupils born outside the country of the test were more evenly distributed among schools in countries such as Luxembourg and Portugal and less evenly distributed in Spain, Finland and Italy. However, the proportion of these pupils varies across the different countries with Luxembourg's sample comprising approximately 18% of these pupils, compared with 2% and 2.4% in Italy and Spain respectively. Despite the strongly compositional invariant nature of the Segregation Index (S), with such a small proportion of the sample in Italy and Spain originating from outside the country of the test, it is unsurprising that they are less likely to be distributed evenly among the school population.

It is also worth examining the relationship between particular groups and academic outcomes (in the case of PISA, this would be performance on the reading test). This is particularly pertinent in the light of current 'moral panics' surrounding the apparent 'underachievement' of boys in British schools—and their performance in English and literacy-based examinations in particular (Gorard *et al.*, 2001b). This analysis also has implications for issues of equity or equality in examination outcomes, particularly with regard to the achievement of pupils from poorer homes. The results in Table 5 suggest that the relative low achievement of boys in literacy examinations is not confined to the UK. Indeed the pattern is repeated across all EU countries with a small proportionate difference in favour of the girls in each case. There appears to be no link with single-sex provision (or indeed any of the measures in Tables 2 and 3). Ireland, with the most sex-segregated school system, has a below EU-average difference in performance by sex. Finland, with one of the largest differences in performance by sex, has no single-sex provision, but nor does

Table 5. Mean reading score by sex

Country	Female	Male	Difference
Austria	516	482	3.4
Belgium	532	501	3.0
Denmark	511	486	2.5
Finland	573	522	4.7
France	517	488	2.9
Germany	514	482	3.2
Greece	492	454	4.0
Ireland	542	513	2.7
Italy	506	471	3.6
Luxembourg	457	433	2.7
Netherlands	557	528	2.7
Portugal	488	464	2.5
Spain	506	482	2.4
Sweden	535	498	3.6
UK	539	511	2.7

Denmark with one of the smallest differences (Table 3). The UK, like Ireland, has a below average difference.

Discussion

While more detailed in their treatment of segregation, these findings are in overall agreement with the summary by Hirsch (2002). Nationally comprehensive systems of schools tend to produce narrower social differences in intake and outcomes. Systems with more differentiation lead to greater gaps in attainment between social groups. Finland, for example, has a high average reading score, a small gap between high and low attainers and comprehensive schools and a policy of choice. Germany, on the other hand, has a much lower average reading score, a large difference between high and low attainers, and a tiered system of selective schooling. The UK is currently still in a reasonable comparative position, with a high average reading score, below average differences between high and low attainers, and comprehensive schools with a policy of (limited) choice.

However, not all commentators are aware of this. There is a common crisis account of the position of UK schooling (and there is, perhaps, a tendency for all commentators to decry the position of their own countries). For example, commenting on the same data as Hirsch (2002), Johnson (2002) recently complained that 'British pupils may be among the world's highest achievers, as the recent Organisation for Economic Cooperation and Development's (OECD) PISA study found. But the achievement gap between social classes remains one of the biggest in the world' (p. 23). This represents the view of the Institute for Public Policy Research (IPPR)—an influential centre-left think tank. The introduction of choice policies have, according to this account, led to a greater polarization of results. But as we have seen, this greater polarization by parental occupation does not exist in the UK

(Johnson was, presumably, referring to the 'social gradient'). Something that does not exist cannot, therefore, be the result of choice policies.

An alternative account of the supposed crisis in the UK education system does exist. For example, recent studies into the extent of the socio-economic stratification in all secondary schools in England have suggested that schools are now more mixed than they were a decade ago (Gorard *et al.*, 2002a). There is also little evidence that the era of school choice has allowed less 'successful' schools to slip into a 'spiral of decline' (Gorard *et al.*, 2002b). And, in addition, analyses of individual pupil performance data have called into question the underachievement of large groups of pupils (Smith, 2002). The first implication of this new study is that when governments talk of 'evidence-based policy making' they must be talking about something other than basing new policy on research evidence. While still facing potential problems such as teacher supply and inequitable funding arrangements, on any rational comparison the UK school system is in the healthiest state ever. Raw score indicators of attainment are rising annually, gaps between social groups are reducing, and socio-economic segregation between schools has declined. We do not appear to need yet more major interventions to solve problems that do not exist and that detract from dealing with the problems that do. The move towards increased diversity of schooling (see introduction) therefore presents a very real danger of increasing existing segregation for very little gain.

The nature of the intake to most schools is determined largely by the nature of local housing (i.e. residential segregation leads to school segregation), and the relative success of schools is determined largely by the nature of their intakes. When the link between housing and schools was weakened *for all* families in the 1980s and early 1990s segregation declined to its lowest recorded level. The general return to catchment areas, following the School Standards and Frameworks Act 1997, signalled an increase in segregation. This increase has been worse in areas where the link between housing and schools has been reinstated for most families, but abolished for a few (i.e. those able to attend schools which recruit across and beyond LEAs). Faith-based, specialist, foundation, selective and Welsh-medium schools all have wider catchments than comprehensives, and this is reflected in their relatively privileged intakes (which may also explain their apparent success in examinations). As the proportion of these minority schools increases, segregation is likely to increase as well. Perhaps we *will* then be able to say with justification 'As every international comparison has shown, English schools are more socially differentiated than any others in Europe'.

Notes on contributors

Stephen Gorard is a professor at the Cardiff University School of Social Sciences. His research is focused on issues of equity, especially in educational opportunities and outcomes. Recent project topics include widening participation in learning, the role of technology in lifelong learning, informal learning (www.cf.ac.uk/socsi/ICT), the role of targets, the impact of market forces on schools (www.cf.ac.uk/socsi/markets), underachievement, teacher supply and

retention, and developing international indicators of inequality (www.cf.ac.uk/socsi/equity). He is a director of the research capacity-building support network for the ESRC Teaching and Learning Research Programme (www.cf.ac.uk/socsi/capacity), and the author of over two hundred publications.

Emma Smith is an educational researcher, having previously been an experienced professional teacher in the secondary sector. She has carried out a range of research and evaluation on the subject of teaching, learning and school organization and has recently acted as special adviser to the House of Commons Select Committee on Education and Skills.

Note

1. *The Dissimilarity index*

$$D = 0.5 \times \sum_{i=0}^{k} \left| \left(\frac{A_i}{X}\right) - \left(\frac{B_i}{Y}\right) \right|$$

A_i and B_i are the number of pupils in the minority (10%) and majority (90%) categories respectively within each school. X and Y are the total number of pupils in either category within each country.

The Gini coefficient

$$G = 1 - \sum_{i=0}^{k-1} (y_{i+1} + y_i)(x_{i+1} - x_i)$$

Where y_I is the cumulative proportion of pupils in the lowest 10% group (or those who were born outside the test country or were female, depending on indicator being measured) and x_i is the cumulative proportion of whole population. The Gini Coefficient is based on the Lorenz curve. This is a cumulative frequency curve that compares the 'distribution of a specific variable with the uniform distribution that represents equality' (Castillo-Salgado *et al.*, 2002, p. 1). This 'equality distribution' is represented by a diagonal line, the greater the deviation of the Lorenz curve from this line, the greater the inequality. The Gini Coefficient ranges from 0 to 1, with 0 representing perfect equality and 1 total inequality.

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